

Identifying Cross-origin Resource Status Using Application Cache

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****Author(s):** ****Sangho Lee, Hyungsub Kim, Jong Kim**

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Abstract:

HTML5 Application Cache (AppCache) allows web applications to cache their same- and cross-origin resources in the local storage of a web browser to enable offline access. However, cross-origin resource caching in AppCache can cause security and privacy problems. In this paper, we consider a novel web privacy attack that exploits cross-origin AppCache. Our attack allows a remote web attacker to exploit a victim web browser to exactly identify the status of target URLs: existence, redirection, or error. Especially, our attack can be performed without using client-side scripts, can concurrently identify the status of multiple URLs, and can exactly identify the redirections of target URLs. We further demonstrate advanced attacks that leverage the basic attack to de-anonymize or fingerprint victims. First, we determine the login status of a victim web browser by identifying URL redirections or errors due to absent or erroneous login information.

Second, we probe internal web servers located in the local network of a victim web browser by identifying URL existence. We also suggest effective countermeasures to mitigate the proposed attacks.

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